

Used Language: Python3

Segment : ACK SYN FIN DATA RESET

Action: Send Segment Receive Segment

Object: sender receiver

To do a classification, it will be clear through all the way.

Organisation: receiver.py sender.py utility.py const.py

receiver.py and sender.py are the files which run, utility.py provide function like creating segment and creating segment, const.py provide constant numbers.

Program Design

Just use class for sequence number while other operations are finished in oriented procession. It will go fluently. With using sleep function for the maximum 3 resend time, it doesn't need timer which it will be used often if with OOP. And use two function in main program, one is "rece" and the other is "send". All the left are helper function.

In summary, one class and two function, form a forward straight and readable program.

Data Structure

In sender side:

Use 'result' key to record if has received or not.

Use 'send_index' to record the seq.

Use 'time_stamp' to record exact time.

Use 'data_len' to record length of data.

Use 'ack_received' to record how many times the ack has been received

In receiver side:

Just set a dict to record the sequence and message that will be sorted and written into file.

Operation

It is the route how the data will generate from left to right.

And some dict or map-relation is at the bottom

Name	Source
max_win	input
rto	input
receiver_port	input-->IP--> receiver_a ddr
FileToSend	input--> send_all _messag FileToSen e--> N = window_d
sender_log	input d → MSS→ pkts → len(pkts) → ata
sender	Create socket Object Exactly
sender_start_time	time.time()
summary_dict	log final part
ISN	random
lef, rig =0 ,0	Windows Tag
summary _dict 'Data_Se 'Amout_D g_Recei 'Data_Se 'ACK_Seg_ ata_No_R ved_No_ g_Resent Received_R peat' Repeat' _Repeat' epeat' window_d ata 'send_in 'time_st 'ack_recei 'result' dex' amp' 'data_len': ved' T F 0 0 DATA_N ACK_N SYN_N RESET_N UM UM UM FIN_NUM UM 0 1 2 3 4	

Trades_offs

Using thread can not be ignore in sender side .However , I don not use thread in receiver because there is no necessity to send ACK in delay which is said in assignment.But it scasifice some accuracy in time.